

Invited talk · Vermeer: Autoregressive generative modeling of microscopy predicts protein localization

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Understanding how proteins localize inside of cells is critical to understanding how cells regulate themselves by moving around proteins, and how these processes are disrupted in disease. Towards characterizing this, systematic microscopy datasets have emerged imaging almost every protein in an organism across cell types or perturbations. However, even high-throughput technologies struggle to fully cover the possible space of experiments in this data collection. In this talk, I will show that generative AI models can help prioritize data collection by simulating unseen experiments to predict which ones will possibly yield informative outcomes. I will introduce our work in Vermeer, which autoregressively generates microscopy images showing how proteins express in cells, predicting what proteins may be differentially regulated across different cell types and how alterations to protein sequence modify this behavior. I will show that Vermeer can accurately simulate the impacts of alterations to protein sequence on localization within cells, providing a basis for discovering novel mediators of localization within the proteome.